

AMAN ADHAV

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Software Engineer | Distributed Systems and Backend | 1st of 50 Teams, Amazon Kiro Challenge | Available Jan 2027

EDUCATION

Arizona State University

Bachelor of Science in Computer Science, GPA 3.83/4.0

Tempe, AZ

Expected Graduation: Dec 2026

Certifications: AWS Certified AI Practitioner, AWS Certified Cloud Practitioner | AWS Developer Associate (in progress)

EXPERIENCE

ASU Enterprise Technology

Software Engineer, Integration & APIs Team

Scottsdale, AZ

Mar 2026 – Present

- Consolidated 21 REST endpoints onto a central data platform, retiring direct SQL access to a legacy system of record and an interim integration layer, building each endpoint across 11 layers (extraction DAO, sync unit, DynamoDB store, REST resource, CDC handler, stream processor, Terraform).
- Authored the platform's OpenAPI 3.0 specification by inventorying every SQL query across four repositories and mapping the upstream table dependency graph; wrote the team's dataset-onboarding guide from the migration.
- Cut event-source onboarding 12x, from 4 hours to 20 minutes, by automating Jira, Terraform, and GitHub PR steps into one approval-gated agent workflow.
- Recovered a customer-facing chatbot from full production outage in under 3 hours by tracing failures to revoked MuleSoft credentials, then shipped a feature-flagged whitelist pre-filter backed by 32 Vitest tests to prevent recurrence.
- Enabled automated end-to-end testing on a legacy app by building a mock authentication Lambda with isolated test personas, reusing gated dev infrastructure instead of provisioning a new AWS environment and database cluster.
- Standardized CI across 12 repositories by authoring reusable GitHub Actions and Jenkins workflows with SonarQube quality gates, replacing per-team setup with one include.

Arizona State University

Undergraduate Teaching Assistant, Calculus & Linear Algebra

Tempe, AZ

Aug 2024 – Mar 2026

Rachita Infotech

Software Engineering Intern

Pune, India

May 2024 – Aug 2024

- Improved internal project-tracking workflows ~15% and raised Jest test coverage from ~60% to ~75% by building a full-stack React/Node/Express app with REST APIs over MongoDB, containerized in Docker and deployed to AWS (EC2, S3).

PROJECTS

AllVoice, Amazon Kiro Challenge Winner | TypeScript, React, Chrome MV3

github.com/amanadhav/AllVoice

- Won Grand Prize and Best in Build (\$6,000, 1st of 50 ASU teams) by architecting a 6-stage voice-to-action pipeline that lets blind users navigate the web hands-free.
- Blocked unsafe browser actions pre-execution via a deterministic ethics gate enforcing 5 privacy rules, verified by 103 automated tests; presented the architecture on Amazon Kiro's official developer livestream.

TraderAI, Self-Hosted Market Analysis Platform | Python, FastAPI, React 19, SQLite

github.com/amanadhav/traderai

- Built a multi-service analysis platform behind 306 hermetic pytest tests and CI on every push, spanning a FastAPI backend, React dashboard, SQLite signal store, and a scheduled daily pipeline.
- Kept the system running through upstream outages with a rate-limit-aware provider fallback chain, file-locked portfolio writes, and token-authenticated mutation endpoints using constant-time comparison.

ClassQ, Distributed Course Registration System | Python, FastAPI, Redis, Terraform

github.com/amanadhav/ClassQ

- Eliminated double-enrollment under 500 concurrent requests using atomic Redis Lua seat allocation, proven by a 500-bot chaos harness asserting zero oversold sections.
- Decoupled event delivery from registration via a transactional outbox and BFS prerequisite-DAG validation, deployed on Terraform-provisioned ECS Fargate.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash, C++

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, SQLite

Libraries & Frameworks: FastAPI, Flask, Node.js, Express, React, Next.js, GraphQL, WebSockets, pytest, Jest, Vitest

Cloud Infrastructure & Deployment: AWS (Lambda, EventBridge, EC2, S3, RDS, ECS Fargate, ElastiCache, Athena, IAM), Terraform, Docker, Jenkins, GitHub Actions, Datadog, Git, Linux